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MAROUN SEMAAN FACULTY OF
ENGINEERING & ARCHITECTURE

VIPP 201A · PROJECT RECAP

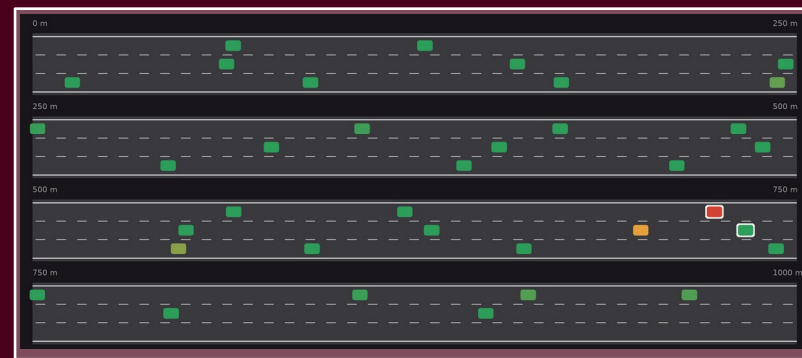
Prediction of Driver Aggressiveness Level via Driving Behavior

From a closed-form index to a sensor-noise-validated, ROS2-integrated system

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Milestone 1 to September 2026 · American University of Beirut



Every car scored live in SUMO: green = calm, red = aggressive



Conservative



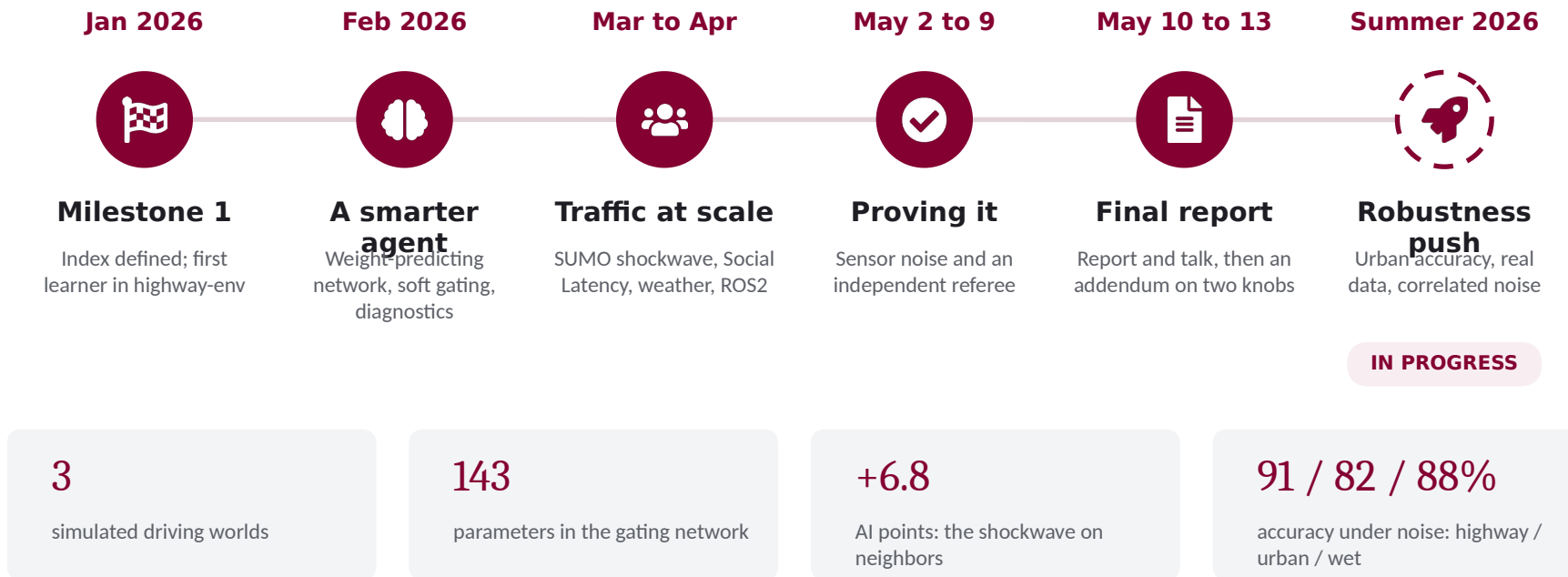
Normal



Aggressive

The journey so far

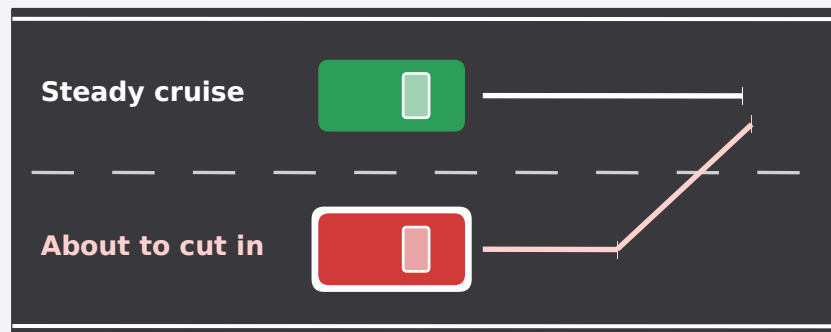
Six stages, one question: can a simple, readable score capture how aggressively a car is driven?



Why predict driver aggressiveness?

Two cars can share the same position and speed, yet be about to do very different things.

Same position, same speed



A tracker sees two identical boxes. The planner needs to know which one to give space to.

Today's AV stacks

Track position and velocity, keep a fixed safety buffer, react after something happens. Every car is a moving obstacle.

This project

Estimate a hidden variable, the Aggressiveness Index (0 to 100), for every nearby car. Anticipate intent and publish the score so a planner can use it.

Goal: an interpretable number a motion planner can trust.

Vocabulary in one slide

The terms that come up again and again in this recap.



Aggressiveness Index (AI)

A 0 to 100 score of how aggressively a car is driven, computed from its motion.



Normalization

Rescaling each raw measurement (km/h, meters) to a 0 to 1 scale so they can be added.



Regime

The driving context: highway, urban, or low-friction weather. Each has its own weights.



Agent

The learning program that makes predictions or decisions inside the simulation.



Reinforcement learning

Learning by trial and error: behavior that earns more reward gets repeated.



Reward

The number the agent tries to maximize. Here, mostly 1 minus its own AI.



Simulator

Software that generates virtual traffic: highway-env early on, SUMO later.



Ground truth

The reference answer used to grade predictions. It must not share the agent's inputs.

The Aggressiveness Index in one picture

Four measurements, each scaled to 0 to 1, blended by weights that sum to 1, then multiplied by 100.



Speed

v / v_{ref} $v_{ref} = 38 \text{ m/s on highway}$



Acceleration

$|a| / 5 \text{ m/s}^2$



Proximity

$1 - \text{gap} / 50 \text{ m}$ *closer = higher*



Waviness

$|\text{lateral drift}| / 2 \text{ m}$

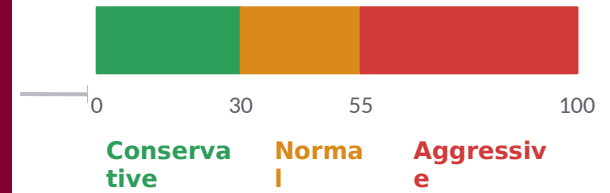
each clipped to [0, 1]

Weighted sum $\times 100$

$$AI = 100 \times (w1 \cdot N_{speed} + w2 \cdot N_{accel} + w3 \cdot N_{prox} + w4 \cdot N_{wave})$$

$$w1 + w2 + w3 + w4 = 1$$

Score 0 to 100 ⁶³



Worked example (highway weights): 30 m/s, 1 m/s², 10 m gap, 0.2 m drift → 0.79, 0.20, 0.80, 0.10 → AI = 63 → **Aggressive**



One formula, three driving worlds

Each regime changes the weights, because the strongest sign of aggression depends on the setting.



Highway

Proximity leads (0.45): tailgating at speed is the clearest single signal.



Urban

Waviness rises (0.30): lane changes and steering jerk matter in stop-and-go traffic.



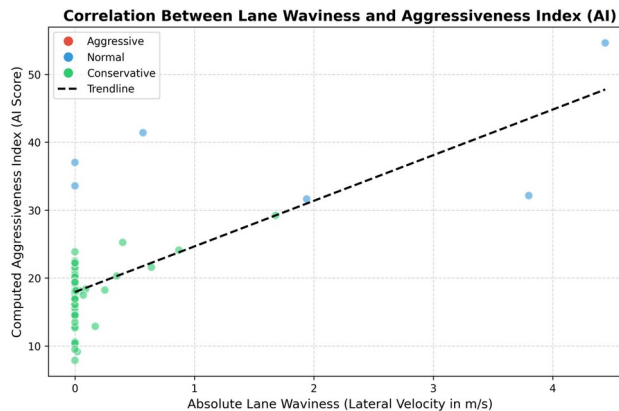
Weather (low friction)

Acceleration leads (0.40): hard throttle on a slippery road is the biggest risk.

Milestone 1 also sketched richer regime-specific signals (time headway, occlusion-aware safe speed, friction-circle use) for future simulators.

Milestone 1: first simulation, first learner

A Gymnasium highway-env world sampled at 15 Hz produced the first datasets and the first correlation.



First finding

The AI score climbs with lane waviness, so lateral instability became a core input.

Also seen

Proximity was the most volatile input, causing spikes; waviness gave a steadier baseline.



First learner

A Q-learner with a 10×3 table: AI bucket (0 to 9) $\rightarrow$ label. Reported near 100% accuracy.



KEY TERM

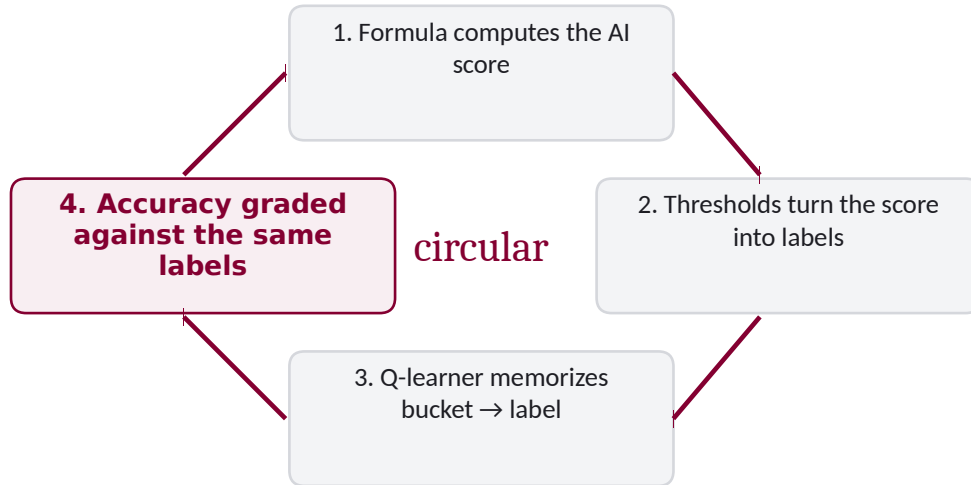
Q-learning: an RL method that keeps a score for every (state, action) pair in a table and nudges it after each reward.

highway-env intersection run (click to play)



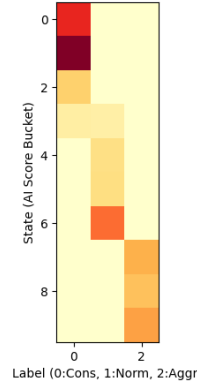
The catch: a perfect score that meant nothing

Dr. Daher's January feedback: the near 100% accuracy was a tautology.



Spring 2026 goal: make learning actually mean something.

Q-Table: Behavioral Probability Mapping



The Milestone 1 Q-table is a staircase that mirrors the threshold rule.

The agent also only saw the trajectories it chose itself.



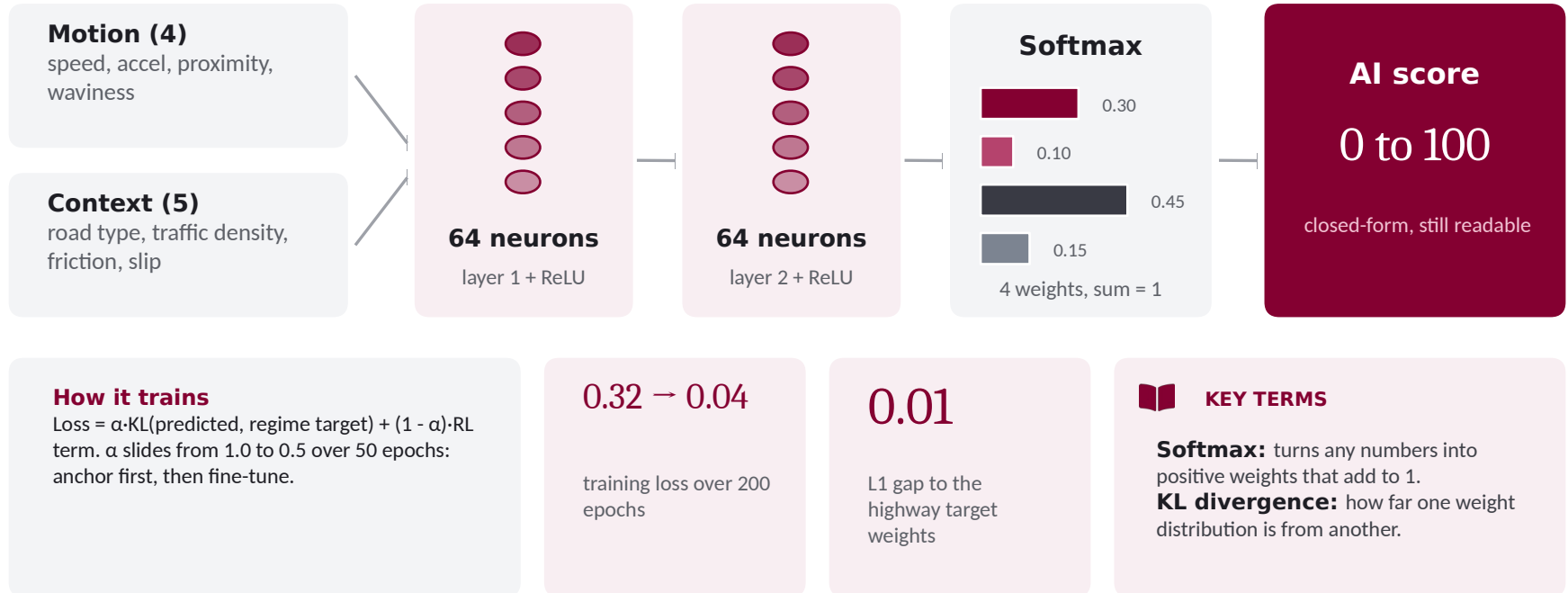
KEY TERM

Tautology: a result that is true by construction, so it carries no evidence.



A new brain: predict the weights, not the label

The DynamicWeightAgent reads motion plus context and outputs the four AI weights. The score stays readable.





Soft switching between regimes

Traffic does not jump from highway to city in one step, so the weights should not either.

How it works

1. A tiny network reads the context and outputs three mixing shares (highway, urban, weather).
2. Final weights = shares $\times$ three learnable regime weight vectors, a smooth blend.

143

parameters in
total

0.05

KL gap to the
targets

0.18

KL with context
zeroed



KEY TERM

Ablation: remove one part and retrain. With context zeroed, the gate collapsed to highway weights, so the context really carries the signal.

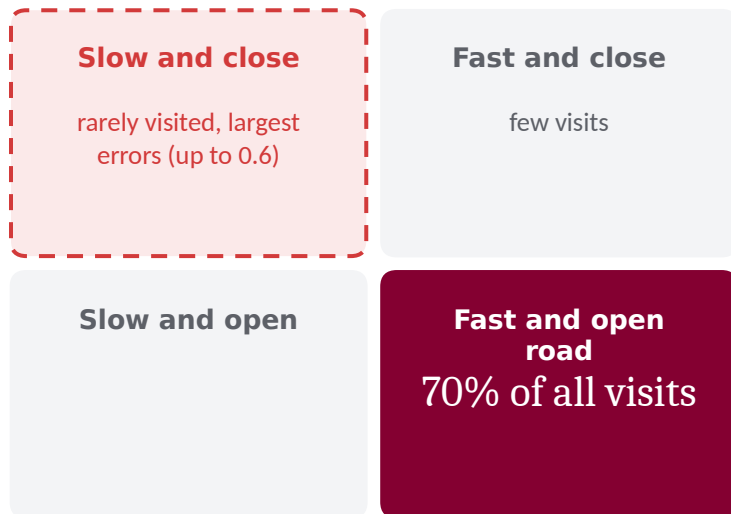
Schematic of the measured behavior: about 12 simulation steps, or 0.8 s at 15 Hz.



Diagnostics: the old learner was hiding something

Week 5 added two instruments to the Q-learner: Bellman-error logging and state-visitation heatmaps.

p
r
o
x
i
m
i
t
y



speed →



Bellman error

How surprised the learner is after each update: $\delta = \text{target} - Q(s, a)$. 95% of values sit below 0.05, but a long tail reaches 0.6, all in one corner.



State visitation

A 32×32 count of where the agent spends its time over 1,000 episodes. Almost no visits where the errors are worst.

Reading: the learner avoided exactly the states where it would be wrong, so its clean accuracy curve was an artifact. This is the conservative-bias problem.



The fix: reward curiosity

A count-based bonus pays the agent for visiting rarely seen states (Bellemare et al., 2016).

$$R_{\text{total}}(s) = R_{\text{env}}(s) + \beta / \sqrt{N(s) + 1}$$

$N(s)$ = number of visits to state s . The bonus fades as a state becomes familiar.

-50%

Bellman error in the
hard corner (0.18 →
0.09)

70 →

40%
share of visits in the
easy corner

+22 pts

accuracy on the rare
Conservative class



KEY TERM

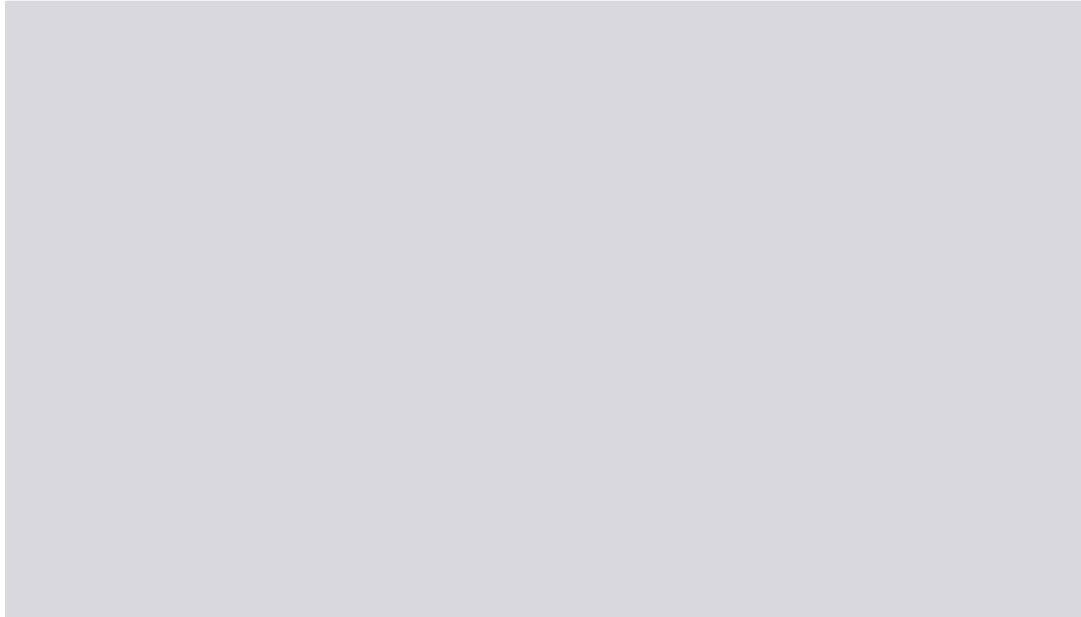
Intrinsic motivation: a reward the agent gives itself for novelty, added on top of the environment's reward.

$\beta = 0.1$ came from a 5-point sweep; the bonus fades out by episode 80. It is re-derived with units in the addendum.



Scaling up: from one car to a whole motorway

SUMO simulates hundreds of cars with realistic car-following and lane changes, controlled live through TraCI.



SUMO motorway rebuilt with the scenario's settings (3 lanes, 1 km, 5% aggressive). Color = live AI. Click to play.

Three scenarios

-  3-lane motorway, 1 km
-  4-arm signalized intersection
-  Low-friction motorway, $\mu = 0.4$

Three driver types

-  Aggressive: 0.5 m min gap, accel 4.0, impatience 1
-  Normal: SUMO defaults
-  Conservative: 4.0 m min gap, impatience 0

KEY TERMS

- IDM:** car-following model (braking and speeding up behind a leader)
- MOBIL:** lane-change model
- TraCI:** live control link into SUMO



Finding 1: aggressive drivers make neighbors worse

5% aggressive drivers mixed into normal traffic vs an all-normal reference (about 600 vehicles, 5 minutes).

+6.8

AI points added to normal drivers, just by sharing the road

Normal drivers: 26.3 in the reference run vs 33.1 with 5% aggressive drivers.

t = 9.8

two-sample t-test

p < 1e-15

p-value: not a fluke

d ≈ 1.5

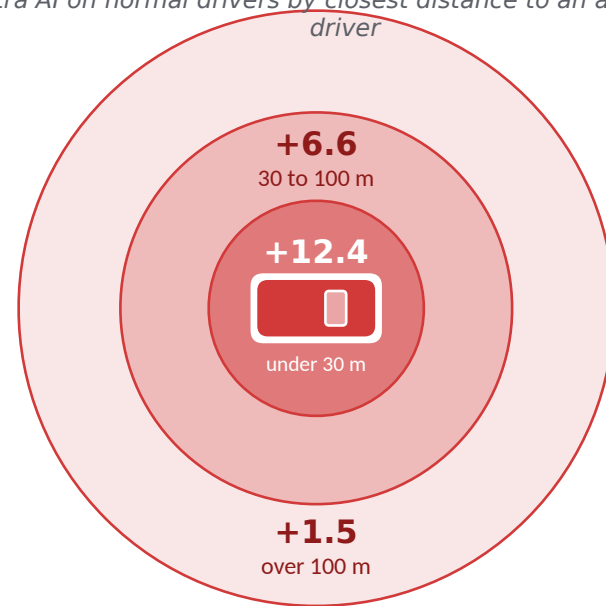
Cohen's d: a large effect



KEY TERM

Shockwave: a disturbance (hard brake, cut-in) that spreads backward through traffic and forces others to react.

Extra AI on normal drivers by closest distance to an aggressive driver





Finding 2: the damage is local and saturates fast

The effect fades within two hops, and the first few percent of aggressive drivers do most of the harm.



KEY TERM

Hop: steps in a network linking cars that came within 30 m of each other.
Hop 1 is a direct neighbor of an aggressive driver.

$$AI(f) = 26.3 + 12.5 \cdot (1 - e^{-(f / 0.06)}), \quad R^2 = 0.998$$

63% of the maximum damage is reached at only 6% aggressive drivers; 95% at 18%.



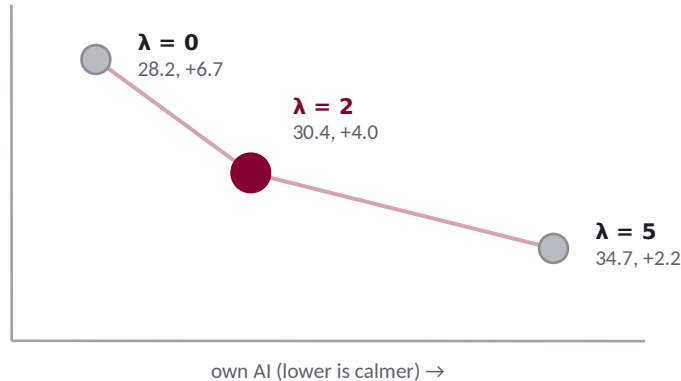
Mitigation: a reward that cares about neighbors

An agent that only minimizes its own AI can panic-brake and start the very shockwave the project detects.

$$R_{\text{total}} = (1 - \text{AI}_{\text{self}}) - \lambda \cdot \max(0, \Delta \text{AI}_{\text{neighbors}})$$

Social Latency penalty. Only harm the agent adds counts: neighbors within 30 m vs a 5 s baseline, clipped at zero.

λ sweep: own calm vs neighbors' calm



+2.8%

average trip time: the price of calmer traffic



KEY TERMS

λ : the trade-off knob
Pareto front: the set of best possible trade-offs



Weather: estimating grip from motion

SUMO has no slip sensor, so slip is estimated from quantities it does report.



Slip estimator

$$\text{slip} \approx |v_{\text{actual}} - v_{\text{intended}}| / v_{\text{actual}}$$

v_{intended} : the speed the car-following model wants on that road.

Dry ($\mu = 0.9$): about 0.01

Wet ($\mu = 0.4$): 0.15 to 0.25 while accelerating

Biased low in steady cruise, but it catches the maneuvers that actually cause slip.

+7.6 vs +6.8 shockwave on wet vs dry roads: aggression costs more when grip is low.



KEY TERM

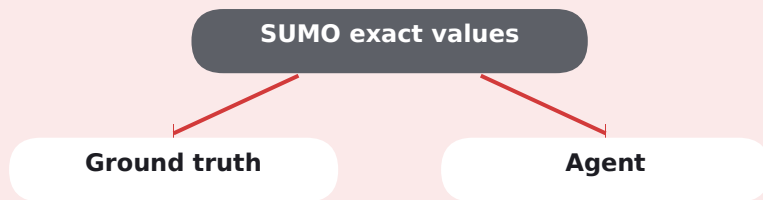
Friction coefficient μ : tire-road grip. Correlations re-tuned the weather weights to 0.10 / 0.40 / 0.15 / 0.35.



May 2: 100% accuracy again, and a bug again

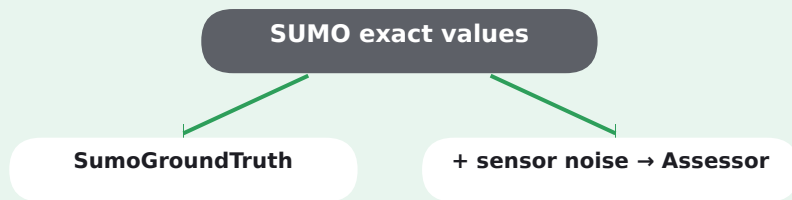
The final smoke test showed perfect agent vs ground-truth agreement everywhere. Perfect means not independent.

BEFORE: information leak



Same inputs, same formula → 100%

AFTER: two independent paths



Agent must estimate from noisy data → honest accuracy

Speed (m/s): true value vs measured with $\sigma = 0.5$ m/s

Noise model

Speed $\sigma = 0.5$ m/s

radar Doppler

Gap $\sigma = 0.8$ m

lidar or radar range

Accel $\sigma = 0.3$ m/s²

fit over noisy speed history



KEY TERM

Gaussian noise: random error on a bell curve; σ sets its spread. Added to inputs, before the formula.



What the leak was hiding

The first honest run scored about 95%, yet labeled zero cars Aggressive. Four root causes, then two cleanups.



Squared speed

01

$(v / 150)^2$ shrinks every value below 1, so speed added at most 30 points.

Fix: linear v / v_{ref} per scenario



Gap to the wrong car

02

Gap was measured to the ego car, not to each car's own leader.

Fix: `traci.getLeader()`



Weather slowed twice

03

Route file scaled speed by 0.7 and the runner cut it again; aggressive cars crawled at about 9 m/s.

Fix: scale once, in the route file



Two-lane intersection arms

04

Aggressive cars overtook, leaving cars with no leader and 150 m gap readings.

Fix: one-lane arms + 8 platoon pairs

Also cleaned up



One reward function

Three scenario branches merged into one, with the regime passed as context. Matches old rewards to floating-point precision.



IDM + MOBIL everywhere

Set explicitly on every vehicle type, no more rule-based stand-ins.



Validated: 91.0%, 81.6%, 88.3%

Agent with sensor noise vs an independent ground truth: 5,437 samples, all three driver classes present in every scenario.



Highway

91.0%

1,328 samples



C 589 · N 717 · A 22



Urban intersection

81.6%

2,284 samples



C 1223 · N 501 · A 560



Weather (rain)

88.3%

1,825 samples



C 833 · N 837 · A 155

Why urban trails. The target was 90 to 95%. At intersection speeds many cars sit close to a category boundary, where a little sensor noise flips the label. The cause is understood, not architectural. Calibration (reference speed, gap, threshold, noise scale per scenario) was locked on May 8.



Lesson, learned twice
100% on anything is a bug signal.



Addendum: giving the hand-picked knobs units

Two numbers came from small grids with no real justification. Both were re-derived so they mean something physical.

BEFORE

$$\beta = 0.1$$

curiosity bonus picked as "not too small, not too large"

AFTER

$$\text{bonus} = c \cdot \sigma(R_{\text{env}}) / \sqrt{(N + 1)}, \quad c = 1$$

A never-seen state is worth one standard deviation of reward. The old $\beta = 0.1$ equals $c \approx 1.1$ ($\sigma \approx 0.09$), and the bonus now shrinks by itself on wet roads ($\sigma \approx 0.06$).

BEFORE

$$\lambda = 2.0$$

Social Latency weight picked from a 5-point grid

AFTER

$$\max E[1 - A_{\text{self}}] \quad \text{subject to} \quad E[\text{harm to neighbors}] \leq \tau = 2$$

τ is in AI points, roughly the harm of a 1% aggressive fleet on the dose-response curve. λ becomes the Lagrange multiplier the optimizer finds (about 2). Report the whole Pareto front.

Also found: ϵ -greedy and curiosity were two knobs doing one job. Next: weight updates by the size of the Bellman error, so exploration switches itself off.



KEY TERM

Lagrange multiplier: the price an optimizer puts on a constraint.



Summer 2026: attacking the limitations

Each weakness named in the final report has a matching work item. Four are in progress.

Urban accuracy is 81.6%



Temporal filter on the noisy inputs (smoother first, Kalman if needed), inside the agent path only, plus urban-specific thresholds. Target: 90% or more with no regressions.

IN PROGRESS

Noise is independent each step



Correlated noise: AR(1) / Ornstein-Uhlenbeck per channel with the same σ values; accuracy vs correlation time.

IN PROGRESS

All traffic is synthetic



Score real trajectories from highD, exiD and NGSIM: check the score distribution and the dose-response shape.

IN PROGRESS

NEXT UP



Paper: IEEE ITSC or T-ITS



CARLA via SUMO co-simulation



Learnable saturation points

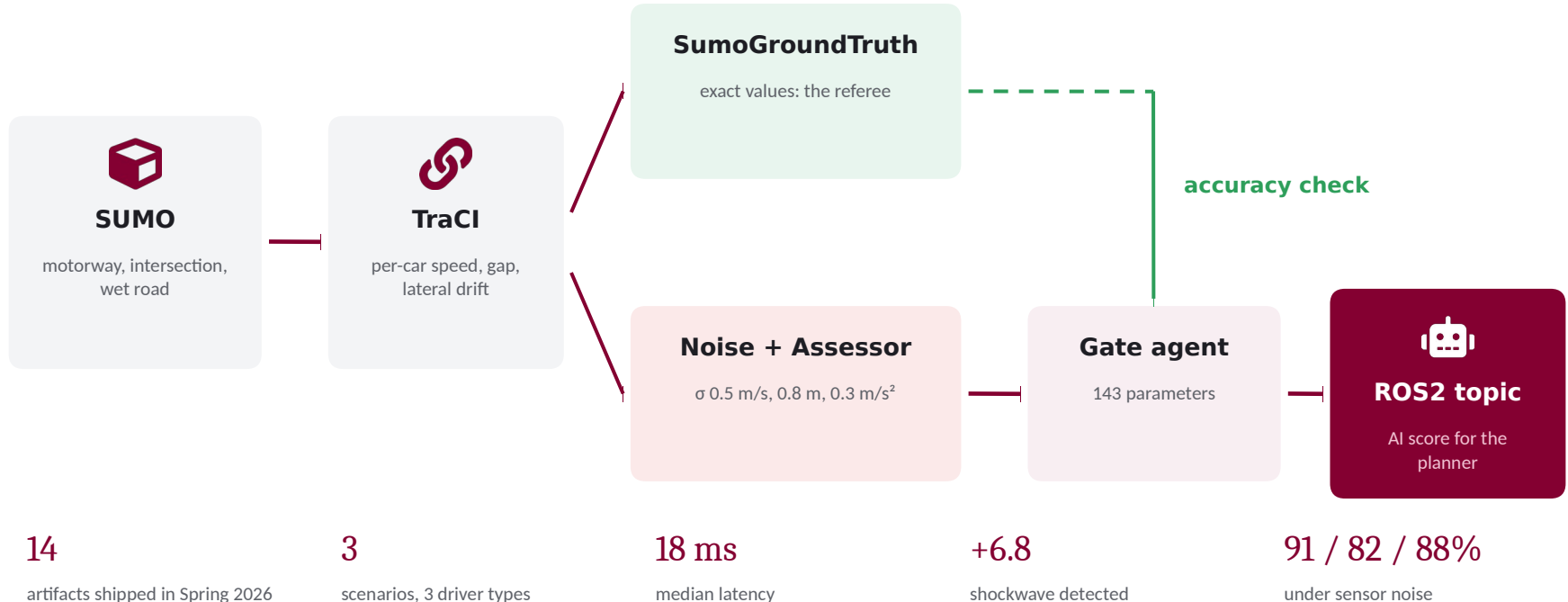


Multi-agent Social Latency RL

Feature	NGSIM	highD	exiD
Source/Location	Stationary cameras (USA)	Drone footage (Germany)	Drone footage (Germany)
Road Geometry	Straight freeways & urban arterials	Straight highway segments	Highway entries, exits & merges
Data Quality	High noise and tracking errors	High precision, low noise	High precision, interactive focus

The system today, end to end

From simulated traffic to a planner-ready topic, with an independent referee checking every prediction.



Three lessons worth keeping



01

100% is a bug signal

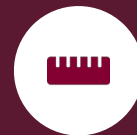
Caught twice: the Milestone 1 Q-learner and the May 2 smoke test. Perfect agreement usually means shared inputs, not a great model.



02

Instruments beat architecture

Bellman-error logs, visitation heatmaps and an independent referee found more problems than any change of model did.



03

Every knob needs units

$c \cdot \sigma(R)$ and a budget in AI points can be defended in front of a panel. $\beta = 0.1$ and $\lambda = 2$ could not.

Glossary


Ablation

Removing one component and retraining to see whether it mattered.

AR(1) noise

Sensor error where each sample partly repeats the previous one, like drift.

Bellman error (δ)

Gap between a learner's value estimate and its updated target.

CMDP

Constrained MDP: maximize reward while keeping a cost under a budget.

Cohen's d

Size of a difference in standard deviations; about 0.8 and above is large.

Dose-response

How an outcome changes as the amount of a cause increases.

Gating network

Small network that blends regime-specific weights smoothly.

Gaussian noise (σ)

Random error on a bell curve with spread σ .

IDM / MOBIL

Standard car-following and lane-changing models used in SUMO.

Kalman filter

Blends a motion model with noisy measurements into a cleaner estimate.

KL divergence

Distance between two probability distributions (here, weight vectors).

Lagrange multiplier

The price an optimizer places on a constraint.

Pareto front

Set of trade-offs where neither goal improves without hurting the other.

Q-learning

Table-based RL that scores every (state, action) pair from rewards.

ROS2 node / topic

A running program / a named channel programs publish and subscribe to.

Slip ratio

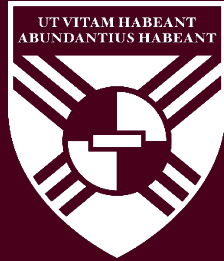
Mismatch between intended and actual motion; rises when grip is lost.

Softmax

Turns any list of numbers into positive values that add to 1.

SUMO / TraCI

Open-source traffic simulator / its live remote-control interface.



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Questions and discussion

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github.com/xHeedz/VIPP-201---SPRING-2025-2026---Dr-Naseem-Daher

Thank you